LESSON PLAN

MOCKUP SQUADRON ENVIRONMENT (MSE) (24)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	10
Total No. of Lesson Plans	21	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Measurement of Breakdown voltage & potential barrier of a diode
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Measurement of Breakdown voltage & potential barrier of a diode.
5. Specific Objectives	24.9.1.1 Trainee will be able to measure various parameters of diode characteristics.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Practical: Measurement of Breakdown voltage and potential barrier using a diode test kit. Safety: Ensure power supply is off before connecting circuit. Instructor supervision: Monitor connections and voltage readings during breakdown test.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a PN junction diode?
(b) Define the term 'potential barrier' in the context of a diode.
(c) What happens to current flow when a diode is reverse biased below breakdown voltage?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Circuit setup for forward bias characteristics • Measurement of potential barrier voltage • Plotting V-I curve for forward region • Identification of knee voltage point • Recording threshold voltage values	L & Ds	PPT, IB	15 min
(b)	• Circuit configuration for reverse bias test • Procedure for increasing reverse voltage • Observation of leakage current increase • Detection of sudden current surge • Recording breakdown voltage value	L & Ds	PPT, PDF	20 min
(c)	• Analysis of forward and reverse V-I curves • Comparison of potential barrier and breakdown voltage • Interpretation of diode characteristic graph • Verification of measured parameters • Documentation of experimental results	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the relationship between potential barrier and forward bias voltage.
(b) Describe the physical mechanism causing diode breakdown at high reverse voltage.
(c) Differentiate between Zener breakdown and avalanche breakdown based on voltage levels.

12. Home Assignment and Follow Up	Prepare a brief report comparing the V-I characteristics of a silicon diode and a germanium diode, highlighting differences in potential barrier and breakdown voltage, to be used in the next lesson on transistor characteristics.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Explanation on the characteristics and testing of a transistor.	Refer to the master syllabus reference list	LP 11

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min